



2005
HIGHER SCHOOL CERTIFICATE
EXAMINATION

Biology

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
- Draw diagrams using pencil
- Board-approved calculators may be used
- Write your Centre Number and Student Number at the top of pages 9, 13, 17 and 21

Total marks – 100

Section I Pages 2–22

75 marks

This section has two parts, Part A and Part B

Part A – 15 marks

- Attempt Questions 1–15
- Allow about 30 minutes for this part

Part B – 60 marks

- Attempt Questions 16–27
- Allow about 1 hour and 45 minutes for this part

Section II Pages 23–33

25 marks

- Attempt ONE question from Questions 28–32
- Allow about 45 minutes for this section

Section I

75 marks

Part A – 15 marks

Attempt Questions 1–15

Allow about 30 minutes for this part

Use the multiple-choice answer sheet.

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

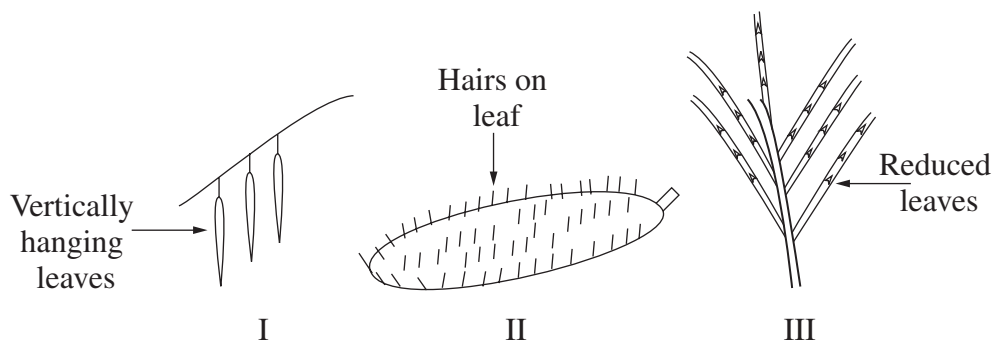
A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word *correct* and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↙

- 1 What is a function of haemoglobin?
- (A) Fights disease
 - (B) Transports oxygen
 - (C) Gives blood its colour
 - (D) Carries salts in the blood
- 2 Which TWO scientists were able to crystallise DNA to develop the first model of DNA?
- (A) Beadle and Tatum
 - (B) Darwin and Wallace
 - (C) Franklin and Wilkins
 - (D) Koch and Pasteur
- 3 Lung cancer is one of the diseases linked to smoking.
- To what category of disease does lung cancer belong?
- (A) Environmental
 - (B) Infectious
 - (C) Nutritional
 - (D) Viral
- 4 Gregor Mendel described the physical traits in pea plants as being controlled by two 'factors'.
- What term is now used to describe the factors Mendel identified?
- (A) Chromosomes
 - (B) Genes
 - (C) Hybrids
 - (D) Polypeptides
- 5 How is the pathogen for malaria transmitted?
- (A) *Anopheles* mosquito
 - (B) Direct human contact
 - (C) Particles in air
 - (D) *Plasmodium falciparum*

- 6 Which of the following best describes the main focus of epidemiology?
- (A) Diseases related to the skin
 - (B) Changes in the characteristics of a species
 - (C) Factors involved in the occurrence, prevalence and spread of disease
 - (D) How the body maintains its functions in response to variations in the environment
- 7 The features of leaves shown in the diagram are adaptations found in some Australian plants.

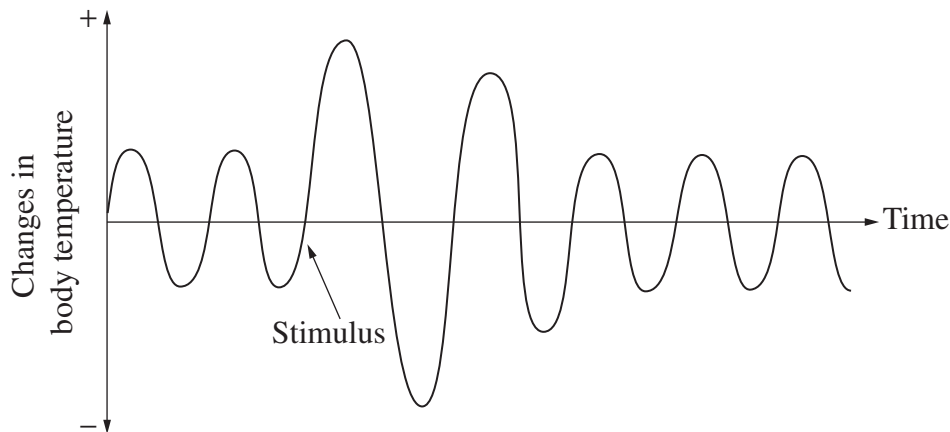


- What is the function of these adaptations?
- (A) Assist in plant growth
 - (B) Allow the transport of water
 - (C) Assist in reducing water loss
 - (D) Reduce the impact of predators
- 8 Recycling of sewage so that it can be used as drinking water is one possible solution to the current water shortages in Australia.

Which of the following would health authorities be concerned about?

- (A) The change to the taste of the water
- (B) The risk of infection from pathogens would increase
- (C) The reluctance of the general public to drink treated sewage
- (D) The effect that water with sediments could have on plumbing systems

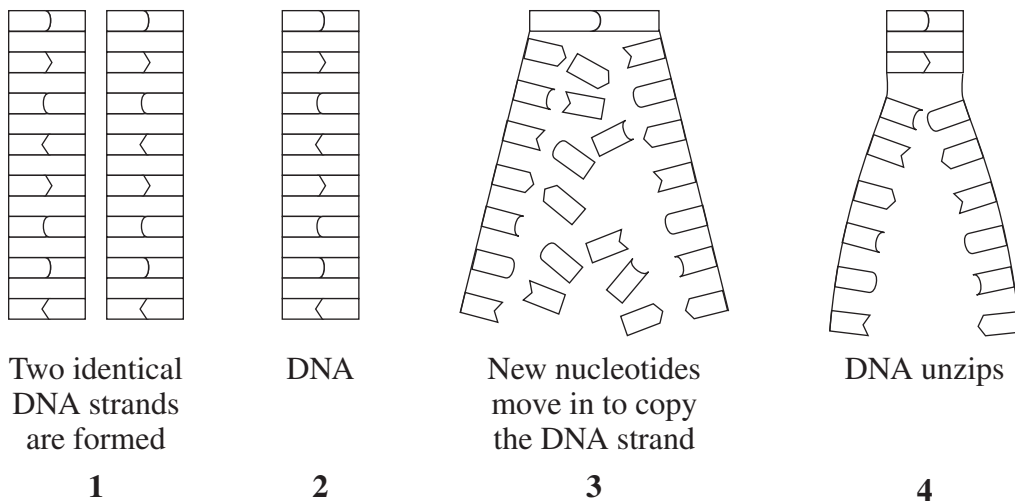
- 9 The graph models the effect of an environmental stimulus on temperature control in mammals.



Reproduced by permission of Oxford University Press Australia from Biology: The Spectrum of Life, 2nd edition, by Peter aubusson et al (c) Oxford University Press, www.oup.com.au

What does this graph represent?

- (A) A reflex action
 - (B) An immune response
 - (C) An impulse pathway
 - (D) A feedback mechanism
- 10 The diagram represents a model of DNA replication. The steps are NOT in the correct order.



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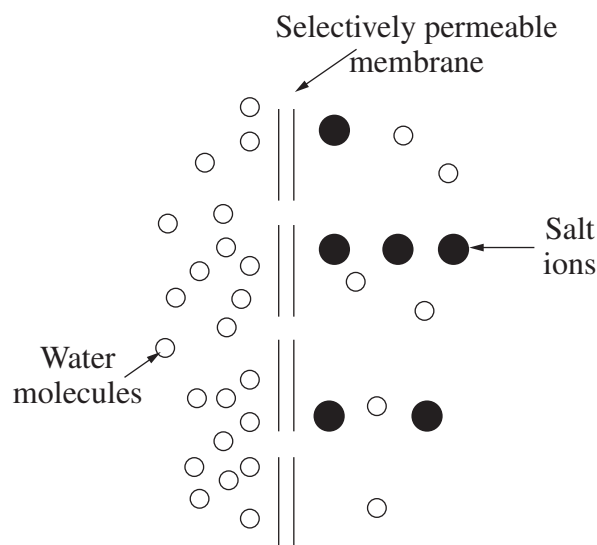
Which of the following is the correct order?

- (A) 1, 2, 3, 4
- (B) 2, 3, 4, 1
- (C) 2, 4, 3, 1
- (D) 4, 3, 2, 1

11 How do vaccines prevent disease?

- (A) Vaccines stop antigens triggering an immune response.
- (B) Vaccines stimulate the production of specific antibodies.
- (C) Vaccines will inhibit the inflammation response in the body.
- (D) Vaccines restrict the vector's ability to inhabit a variety of environments.

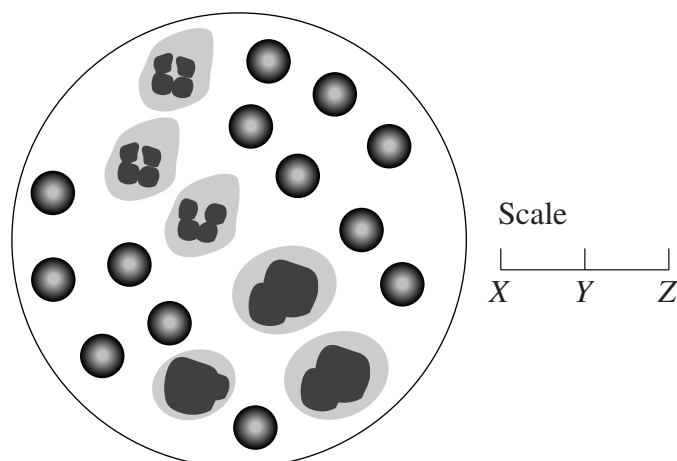
12 The diagram is a model of an important biological process.



What prediction can be made using the model?

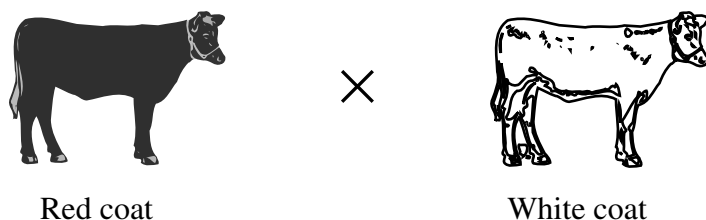
- (A) Salt ions move by osmosis into an area of low solute concentration from an area of high solute concentration.
- (B) Salt ions move by diffusion from an area of high solute concentration to an area of low solute concentration.
- (C) Water molecules move by diffusion from an area of high solute concentration to an area of low solute concentration.
- (D) Water molecules move by osmosis from an area of low solute concentration to an area of high solute concentration.

- 13 The diagram represents red and white blood cells seen under a light microscope.



What do X, Y and Z represent on the scale?

- (A) 0, 10, 20 nm
 (B) 0, 10, 20 μm
 (C) 0, 50, 100 μm
 (D) 0, 50, 100 mm
- 14 A geneticist was studying coat colour in a herd of Shorthorn cattle. She performed the following cross:



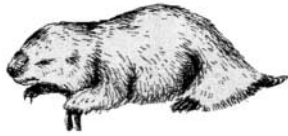
The results of the cross are shown in the table below.

<i>Phenotype</i>	<i>Offspring number</i>
White coat	0
Red coat	0
Red and white coat (roan)	157
Total	157

The roan coat colour was the result of which type of inheritance?

- (A) Co-dominance
 (B) Dominant/recessive
 (C) Heterozygous parents
 (D) Hybridisation

- 15 The two animals pictured are not closely related. They burrow and live most of their life underground.



Australian marsupial mole
(*Notorcytes caurinus*)



European placental mole
(*Talpa europaea*)

Which of the following explains how these two animals can be used to illustrate Darwin/Wallace's theory of evolution by natural selection and isolation?

- (A) They demonstrate comparative anatomy.
- (B) They must have evolved from a common ancestor.
- (C) They demonstrate that an organism can change its characteristics to suit its current environment.
- (D) They have evolved similar structures in response to similar environmental conditions.

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Centre Number

Section I (continued)

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Student Number

Part B – 60 marks

Attempt Questions 16–27

Allow about 1 hour and 45 minutes for this part

Answer the questions in the spaces provided.

Marks

Question 16 (3 marks)

- (a) Define the term
- pathogen*
- .

1

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- (b) Complete the table below to distinguish between bacteria and viruses.

2

	<i>Characteristic/feature of organism</i>
Bacteria	
Virus	

Question 17 (3 marks)

Use an example to outline the role of the nervous system in maintaining homeostasis. **3**

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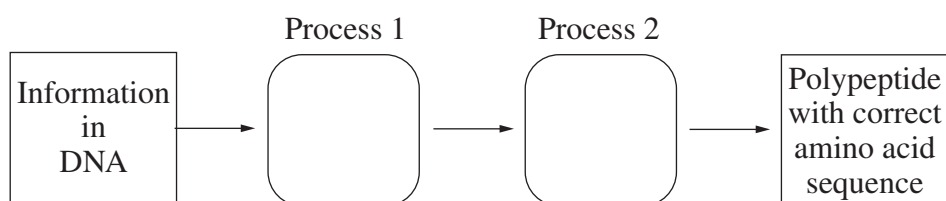
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Question 18 (4 marks)

This flow chart represents a model of polypeptide production.

4



In the table below, name and outline what occurs in Processes 1 and 2.

<i>Name of process (word or phrase)</i>	<i>Outline of process</i>
Process 1 	
Process 2 	

Question 19 (4 marks)

During your study of Biology you have been required to gather and analyse information on an infectious disease.

4

Fill in the table below for ONE infectious disease.

Name of the disease

<i>Feature</i>	<i>Description</i>
Cause	
Symptoms	
Treatment	
Prevention	

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Centre Number

Section I — Part B (continued)

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Student Number

Marks

Question 20 (3 marks)

This table shows the results of the analysis of urine samples of three organisms.

3

<i>Organism</i>	<i>Terrestrial mammal</i>	<i>Freshwater fish</i>	<i>Marine fish</i>
<i>Urine concentration</i>	concentrated	dilute	concentrated
<i>Form of nitrogen waste</i>	urea	ammonia	urea

Using information from the table, explain the differences in urine concentration of these three organisms.

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(a) Why is it necessary for excess carbon dioxide to be removed from blood? **1**

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(b) In what forms is carbon dioxide transported in blood? **2**

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(c) You performed a first-hand investigation to demonstrate the effect of dissolved carbon dioxide on the pH of water. 5

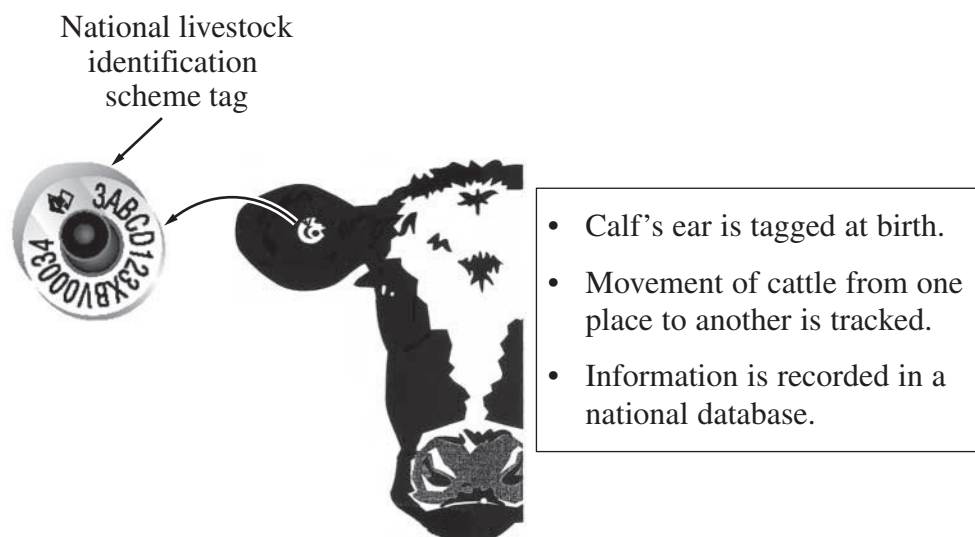
Outline your method, identifying the variables AND account for how you ensured the reliability of your data.

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Question 22 (4 marks)

Lifetime traceable tags as shown in the diagram are now required on all cattle in Australia.

4



Department of Primary Industries, Victoria, 'National Livestock Identification Scheme (NLIS)' on <http://wwdpi.vic.gov.au/dpi>

Discuss the use of this strategy in dealing with diseases.

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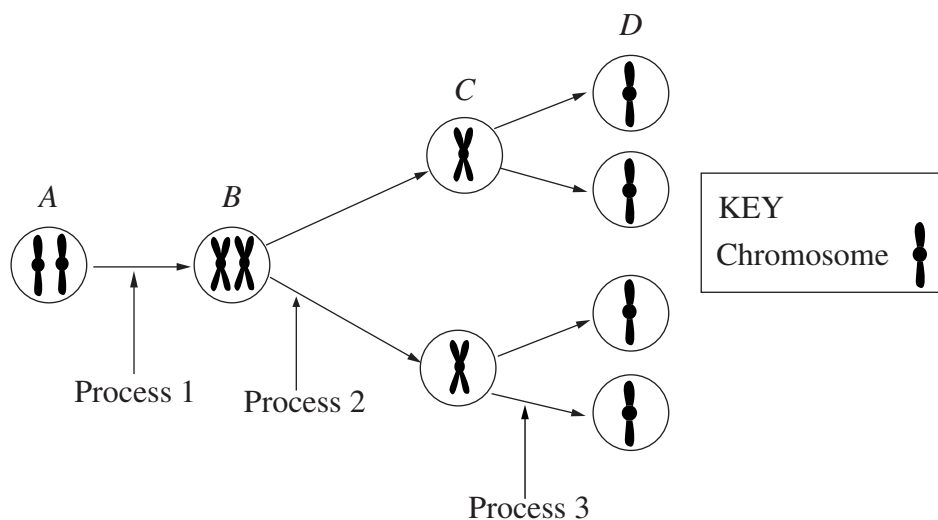
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Question 23 (5 marks)

The flow diagram below is a representation of one pair of homologous chromosomes in a cell during meiosis.



- (a) Identify ONE characteristic of the cells labelled *D*. 1

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- (b) Identify what occurs in Process 1. 1

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- (c) Identify where crossing over would occur, and explain how crossing over affects the inheritance of genes. 3

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Section I – Part B (continued)

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Student Number

Question 24 (6 marks)

(a) Construct a family pedigree to show the inheritance of this sex-linked genetic disorder.

4

Date	Time	Location	Description
2023-10-26	08:00	Forest A	Initial survey of Forest A. No significant findings.
2023-10-26	10:30	Forest B	Survey of Forest B. Observed 12 individuals of species X.
2023-10-26	13:00	Forest A	Re-survey of Forest A. Found 5 new individuals.
2023-10-27	07:30	Forest C	Early morning survey of Forest C. No sightings.
2023-10-27	09:00	Forest B	Survey of Forest B. Observed 15 individuals of species X.
2023-10-27	11:30	Forest A	Survey of Forest A. Found 8 individuals.
2023-10-27	14:00	Forest C	Survey of Forest C. Observed 10 individuals.
2023-10-28	08:30	Forest B	Survey of Forest B. Observed 18 individuals of species X.
2023-10-28	10:00	Forest A	Survey of Forest A. Found 10 individuals.
2023-10-28	12:30	Forest C	Survey of Forest C. Observed 12 individuals.
2023-10-28	15:00	Forest B	Survey of Forest B. Observed 16 individuals of species X.
2023-10-29	07:00	Forest A	Early morning survey of Forest A. No sightings.
2023-10-29	09:30	Forest B	Survey of Forest B. Observed 14 individuals of species X.
2023-10-29	12:00	Forest C	Survey of Forest C. Observed 11 individuals.
2023-10-29	14:30	Forest A	Survey of Forest A. Found 9 individuals.
2023-10-29	17:00	Forest B	Late afternoon survey of Forest B. Observed 13 individuals of species X.
2023-10-30	08:00	Forest C	Survey of Forest C. Observed 10 individuals.
2023-10-30	10:30	Forest A	Survey of Forest A. Found 11 individuals.
2023-10-30	13:00	Forest B	Survey of Forest B. Observed 17 individuals of species X.
2023-10-30	15:30	Forest C	Survey of Forest C. Observed 13 individuals.
2023-10-30	18:00	Forest A	Late afternoon survey of Forest A. Found 12 individuals.
2023-10-31	07:30	Forest B	Early morning survey of Forest B. Observed 14 individuals of species X.
2023-10-31	09:00	Forest C	Survey of Forest C. Observed 11 individuals.
2023-10-31	11:30	Forest A	Survey of Forest A. Found 10 individuals.
2023-10-31	14:00	Forest B	Survey of Forest B. Observed 16 individuals of species X.
2023-10-31	16:30	Forest C	Late afternoon survey of Forest C. Observed 12 individuals.
2023-10-31	19:00	Forest A	Nighttime survey of Forest A. No sightings.
2023-11-01	08:30	Forest B	Survey of Forest B. Observed 15 individuals of species X.
2023-11-01	11:00	Forest C	Survey of Forest C. Observed 10 individuals.
2023-11-01	13:30	Forest A	Survey of Forest A. Found 11 individuals.
2023-11-01			

(b) Predict whether Jack will be colour blind. Justify your answer.

2

[illegible]

Question 25 (8 marks)

Use the information provided to answer the following questions.

The Locust Plague

The Australian plague locust is a damaging, recurring, insect pest of pastures and crops throughout south-eastern Australia. The greatest activity is when locusts feed and move, usually from spring to autumn. Locust occurrence depends entirely on weather and feed conditions. Locusts reproduce rapidly, migrate hundreds of kilometres overnight and destroy large areas of pastures and crops.

Adapted from David Croft, 'Plague Locusts and Spring', IREC Farmers' Newsletter, No. 167, Winter 2004, p18.

- (a) Identify ONE appropriate method to estimate how widespread a locust infestation might be. **1**

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- (b) State ONE other valid piece of information that could be collected about locusts, and identify how this information is useful in developing a method for controlling the pest. **2**

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Question 25 continues on page 19

Pesticide to fight locust plague

Fungus to fight locust plague

Adapted from Steve Letts, 'Australia's largest locust plague', 12 November 2005, Landline, ABC Online, by permission of ABC Content Sales.

- 5

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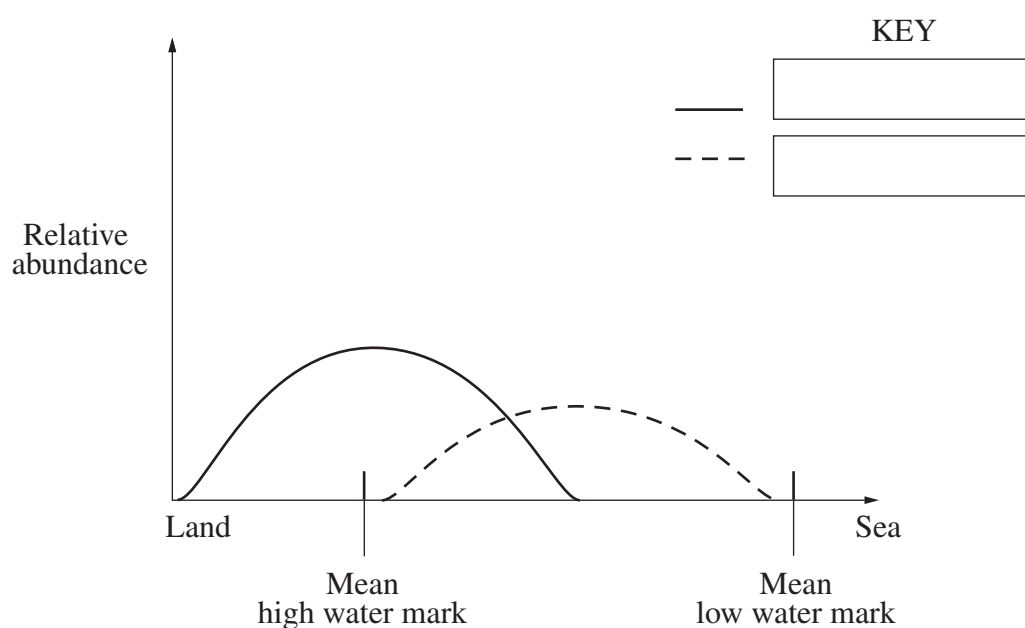
Question 26 (4 marks)

The graph illustrates the relative distribution of two species of mangrove growing on the margin of an estuary.

4

The main characteristics of the leaves of these mangrove plants are:

<i>Mangrove</i>	<i>Characteristics</i>
Species A	Leaves covered with salt crystals
Species B	Leaves with thick waxy cuticles



Complete the KEY in the boxes provided above, to identify which line represents Species A and which line represents Species B. Justify your answer.

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Centre Number

Section I — Part B (continued)

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Student Number

Marks**Question 27** (8 marks)

The influenza virus has a high rate of mutation which can lead to changes of the viral surface antigens that contain protein.

8

Analyse the impact of high mutation rates for this virus and the implications for human health.

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Question 27 continues on page 22

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– 22 –

Biology

Section II

25 marks

Attempt ONE question from Questions 28–32

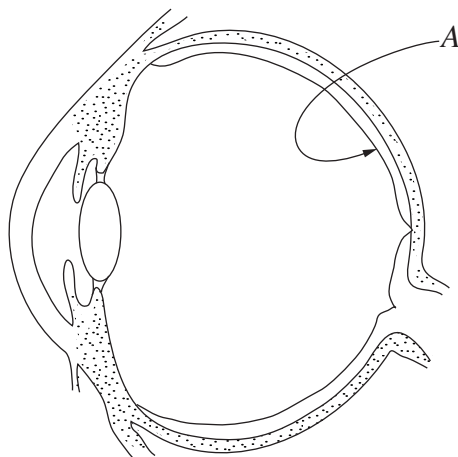
Allow about 45 minutes for this section

Answer the question in a writing booklet. Extra writing booklets are available.

	Pages
Question 28 Communication	24–25
Question 29 Biotechnology	26
Question 30 Genetics: The Code Broken?	27–29
Question 31 The Human Story	30–31
Question 32 Biochemistry	32–33

Question 28 — Communication (25 marks)

- (a) The diagram provided is of the human eye.



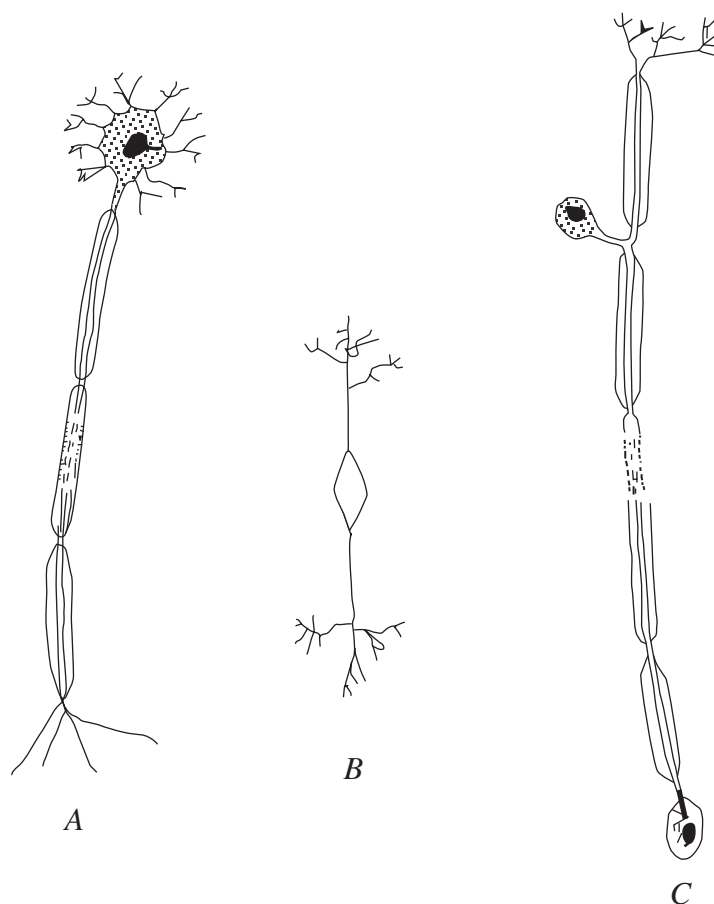
- | | | |
|--|---|----------|
| (i) | Identify the part of the organ labelled A. | 1 |
| (ii) | Using TWO labelled diagrams, show how the eye accommodates the focus of near and far objects. | 3 |
|
(b) Two of your elderly relations have developed loss of sight. One case is due to cataracts and the other is due to myopia. | | |
| (i) | Describe each of these conditions. | 2 |
| (ii) | Compare the technologies available to correct these conditions. | 4 |
|
(c) Hearing aids and cochlear implants are technological advances that have been developed to assist people with hearing difficulties. | | 7 |

Evaluate the benefits and limitations of BOTH technologies.

Question 28 continues on page 25

Question 28 (continued)

(d) The diagram provides a pictorial representation of the structure of neurones.



- | | |
|---|----------|
| (i) From the diagram, identify TWO differences between neurones. | 2 |
| (ii) Assess the use of prepared slides or micrographs when gathering information about the structure of neurones. | 3 |
| (iii) Explain why NOT all stimuli generate an action potential. | 3 |

End of Question 28

Question 29 — Biotechnology (25 marks)

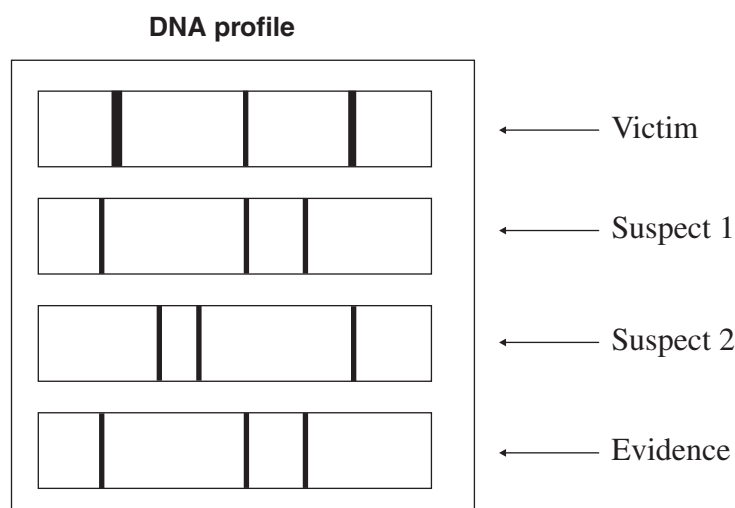
- (a) The equation for fermentation in the production of alcohol is



- (i) What is the name of the product *A* in the above word equation? **1**
- (ii) Name and describe a modern industrial fermentation process that produces large quantities of consistent product. **3**
- (b) (i) Define the term *recombinant DNA technology*, and give an example. **2**
- (ii) Compare recombinant DNA technology and artificial selection. **4**
- (c) Research into biotechnology including medicine, animal biotechnology, and aquaculture, has led to the development of a variety of applications. **7**

Evaluate the limitations and expected benefits of particular applications that you presented in TWO of your case studies.

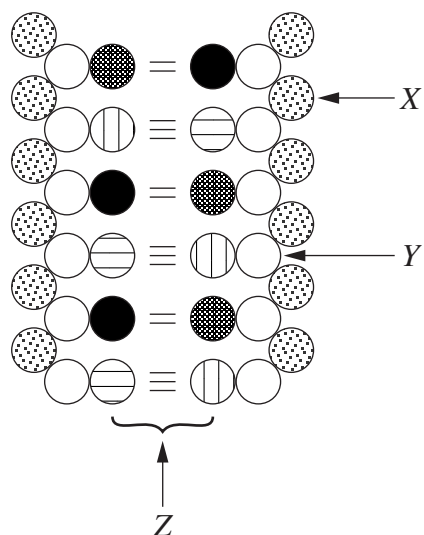
- (d) The diagram represents the DNA profiles from a murder case.



- (i) Justify which one of the suspects could have been the murderer. **2**
- (ii) Explain why a sample would be taken from the victim as well as the suspects. **3**
- (iii) Outline the use of DNA fingerprinting as an investigation tool in forensic cases. **3**

Question 30 — Genetics: The Code Broken? (25 marks)

- (a) During your study of this option you constructed a model of DNA. The diagram represents a model made by a student.

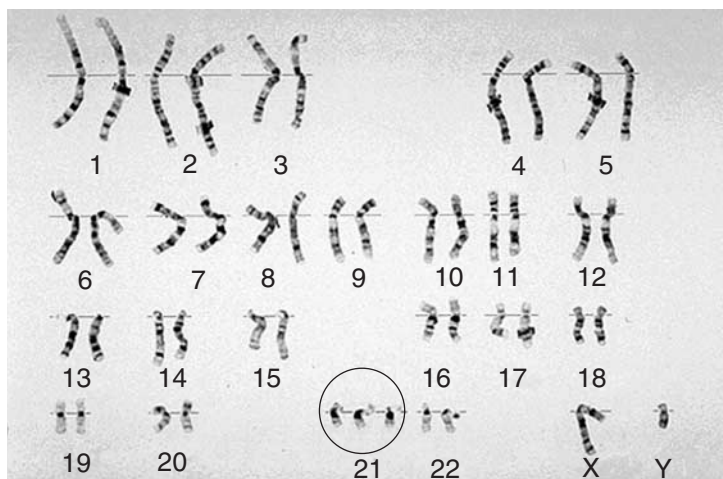


- (i) In this model, name TWO of the parts labelled X, Y and Z. 2
- (ii) Is this a useful model of a DNA molecule? Support your choice with TWO reasons. 2

Question 30 continues on page 28

Question 30 (continued)

- (b) The diagram shows the arrangement of human chromosomes. A chromosomal abnormality is circled.



- (i) Determine how many chromosomes this person has, and name the type of chromosomal disorder. 2
- (ii) Compare the effect of germ line and somatic mutations on a species. 4
- (c) The Human Genome Project has identified the DNA-base sequence of all human chromosomes. A significant amount of work still needs to be done so that this information can be used to benefit humans. 7

Using examples, evaluate the limitations and expected benefits of the Human Genome Project.

Question 30 continues on page 29

Question 30 (continued)

- (d) Indian corn has four grain types that involve combinations of colour (*b*) and appearance (*r*).

Four genes are involved, which are located on two pairs of homologous chromosomes (each gene on a separate chromosome). The alleles are:

B = black ***b*** = yellow ***R*** = round ***r*** = wrinkled

- (i) The table below shows the results of a dihybrid cross between two heterozygous parents, (***BbRr*** × ***BbRr***).

<i>Grain phenotype</i>	Observed number of offspring
Black and round	144
Black and wrinkled	48
Yellow and round	48
Yellow and wrinkled	16
Total	256

Using this data, calculate the ratio of phenotypes in this cross.

1

- (ii) If, however, ***B*** and ***r*** are on one chromosome and ***b*** and ***R*** are on another, draw a Punnett square of a dihybrid cross (***BbRr*** × ***BbRr***). Include the phenotypic and genotypic ratios of the offspring.
- (iii) Explain how cross-breeding experiments can identify the relative positions of linked genes.

3

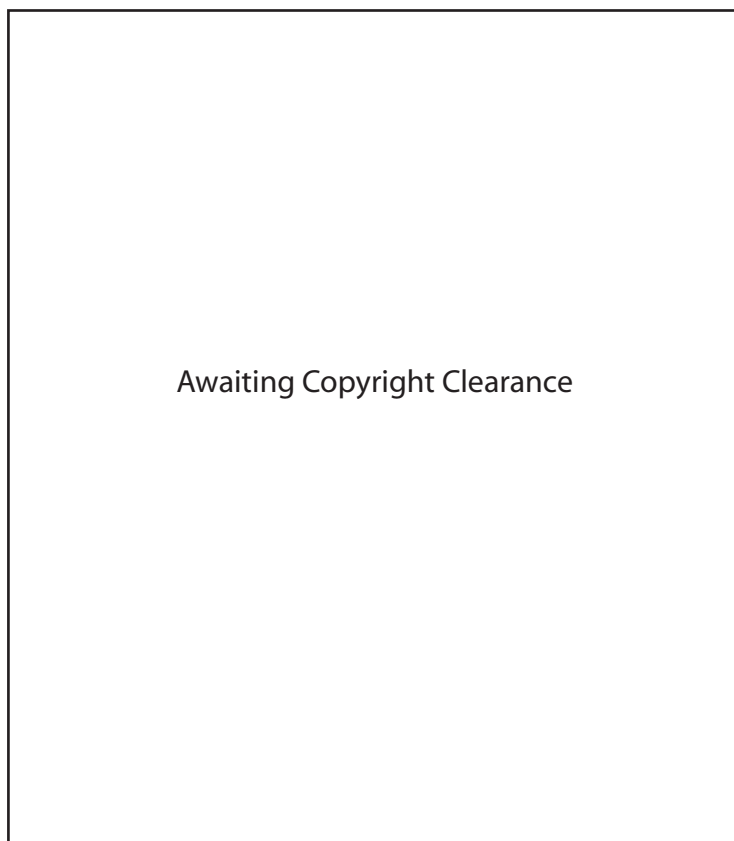
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End of Question 30

Question 31 — The Human Story (25 marks)

- (a) A student was investigating evolutionary relationships between humans and other primates. The diagram models the data collected.

Chromosome 5 in apes and humans



- | | | |
|------|---|---------------------------------|
| (i) | What is this type of analysis called? | 1 |
| (ii) | Identify THREE ways in which this model is useful in showing evolutionary relationships. | 3 |
| (b) | <p>(i) Outline ONE feature for any TWO of the following groups into which humans are classified.</p> <ul style="list-style-type: none"> • primate • hominid • <i>Homo sapiens</i> <p>(ii) Discuss the difficulty of interpreting the past using only fossil records.</p> | <p>2</p> <p>4</p> |

Question 31 continues on page 31

Question 31 (continued)

- (c) Analyse future trends in biological evolution in terms of increased population mobility, modern medicine and genetic engineering. 7
- (d) In 2004 a team of Australian and Indonesian researchers outlined the discovery of the remains of dwarf-sized people who lived on the island of Flores in Central Indonesia. They belong to a species new to science and have been named *Homo floresiensis*. Dating techniques suggest that the remains are up to 95 000 years old.

Homo floresiensis were island people, no more than one-metre tall, with brains the size of newborn modern humans. However, they exhibited behavioural sophistication and intelligence. Their discovery suggests that instead of following a simple evolutionary path, culminating in *Homo sapiens*, (Figure 1) human evolution developed more varied forms than previously thought (Figure 2).

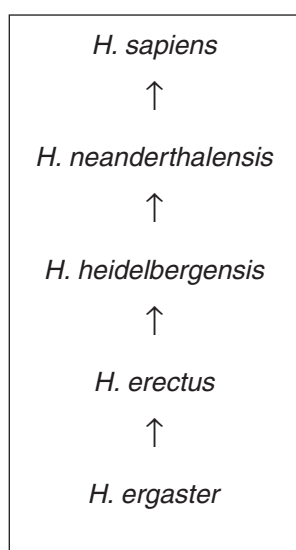


Figure 1

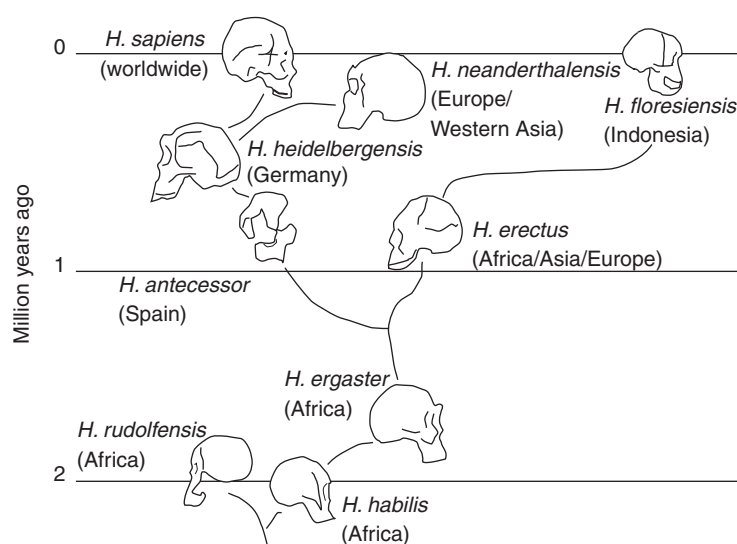


Figure 2

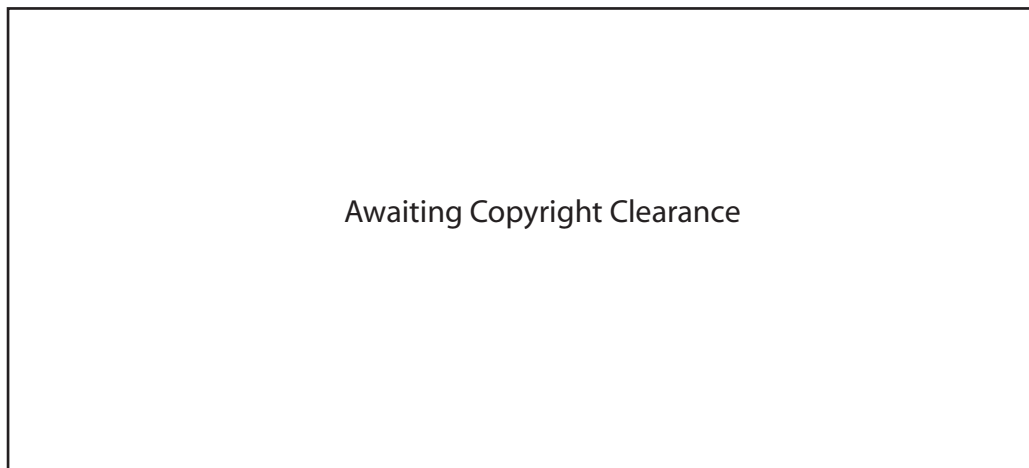
© NewScientist

- (i) Distinguish between relative and absolute dating technique, and outline an absolute dating technique that could be used to date the fossils of *Homo floresiensis*. 3
- (ii) Outline a mechanism that led to the development of *Homo floresiensis* as a distinct species. 2
- (iii) Name the TWO models of human evolution shown in Figure 1 and Figure 2 and compare the evidence that supports them. 3

End of Question 31

Question 32 — Biochemistry (25 marks)

- (a) In the late 1770s the Dutch scientist Jan Ingen-Housz performed a series of experiments shown in the diagrams.



- (i) Name ONE pigment that can be extracted from leaves. 1
- (ii) Identify THREE pieces of information that can be obtained from these experiments. 3
- (b) Two species of plants, X and Y, have been selected for studies on photosynthesis. The plants are two months old, 30 cm tall and 900 g in weight. It is found that their oxygen production varies as shown in the table below.

<i>Time</i> (hour)	<i>Volume of oxygen produced by species X</i> (mL)	<i>Volume of oxygen produced by species Y</i> (mL)
1	2	5
2	4	10
3	6	15
4	8	20
5	10	25

- (i) Based on the information provided, identify TWO microscopic structural differences between plants X and Y. Support your answer using examples. 3
- (ii) You processed information from secondary sources about light absorption and the Calvin cycle. Identify the sites of light absorption and the Calvin cycle and explain their locations. 3

Question 32 continues on page 33

Question 32 (continued)

(c)

Green algae have had their photosynthetic pathways reprogrammed to produce hydrogen which can be used as an alternative energy source.

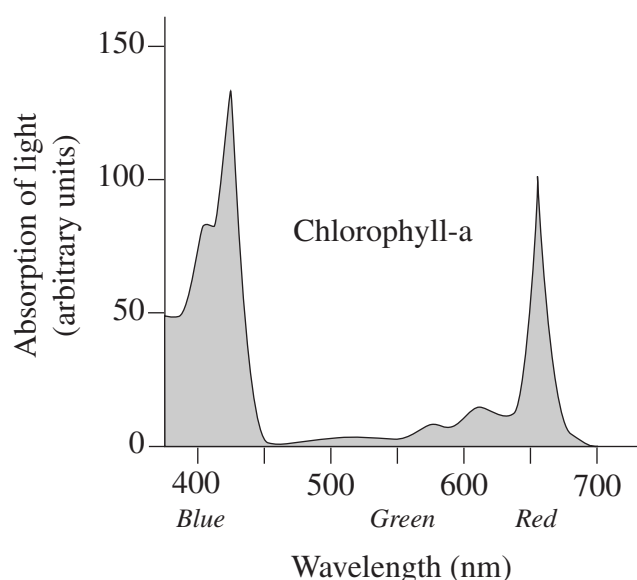
Wired Magazine, Issue 10.04, April 2002

An understanding of photosynthetic biochemical pathways has impacted on the development of new technologies.

Evaluate the potential uses of photosynthesis in replacing at least **THREE** named materials presently obtained from other non-renewable resources.

7

(d) The following figure shows the absorption spectrum of chlorophyll-a.



M Rowland, 1992, Biology, Thomas Nelson and Sons Ltd, Walton-on-Thames, Surrey, p 145.

The owners of a glasshouse are covering the glass with a filter to produce a crop in the shortest possible time.

(i) Explain which filter you would suggest.

3

(ii) Describe photosystems I and II and explain their role in photosynthesis.

5

End of paper

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**2005 HSC Notes from
the Marking Centre
Biology**

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2005 HSC NOTES FROM THE MARKING CENTRE

BIOLOGY

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Biology. It provides comments with regard to responses to the 2005 Higher School Certificate Examination, indicating the quality of candidate responses and highlighting the relative strengths and weaknesses of the candidature in each section and each question.

This document should be read along with the relevant syllabus, the 2005 Higher School Certificate examination, the marking guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Biology.

General Comments

In 2005, 13217 candidates attempted the Biology examination. The most popular electives were Communication (58%) and Genetics: The Code Broken? (20%).

Teachers and candidates should be aware that examiners may write questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course, including the Prescribed Focus Areas. This reflects the fact that the knowledge, understanding and skills developed through the study of discrete sections should accumulate to a more comprehensive understanding than may be described in each section separately. It is important to understand that the Preliminary HSC course is assumed knowledge for the HSC course.

In 2005, at least one question in Section 1 Part B focused on the mandatory skills content in Module 9.1. Candidates who had actively planned and performed practical experiences clearly demonstrated a deeper knowledge and understanding of the content described in this module. There was evidence that some candidates had a very poor knowledge of basic definitions specific to terminology associated with the course.

Overall, the candidate's responses were appropriate and indicated a level of understanding of Biology concepts, appropriate for most HSC candidates. Candidates need to be reminded that the answer space provided and the marks allocated are guides to the maximum length of response required. Similarly, the key word used in the question gives an indication of the depth of the required response. Candidates should use examination time to analyse the question and plan responses carefully, working within that framework to produce clear and concise responses. This may include the use of dot points, diagrams and/or tables, and avoids internal contradictions. This is particularly so in holistic questions which need to be logical and well structured.

Better responses indicate that candidates are following the instructions provided on the examination paper. In these responses, candidates:

- show all working where required by the question
- do not repeat the question as part of the response
- look at the structure of the whole question and note that in some questions the parts follow from each other ie responses in part (a) lead to the required response in part (b) etc

- use appropriate equipment, for example, pencils and a ruler to draw diagrams and graphs. (A clear plastic ruler would aid candidates to plot points that are further from the axes and rule straight lines of best fit.)

The option question is divided into a number of parts: candidates should clearly label each part of the question when writing in their answer booklets. In part (c), many candidates wrote a lot of information that was not relevant to the question. Some responses show evidence of rote learning an anticipated answer based on a single source. These did not address the syllabus content and/or outcomes being assessed and hence did not score full marks. Some candidates are responding to more than one option question: candidates are strongly advised to answer the option they have studied in class.

Section I – Core

Part A – Multiple choice

Question	Correct Response
1	B
2	C
3	A
4	B
5	A
6	C
7	C
8	B

Question	Correct Response
9	D
10	C
11	B
12	D
13	B
14	A
15	D

Part B

General Comments

Overall, the responses indicated that the majority of candidates had a good grasp of biological concepts, appropriate for HSC Biology. Candidates need to be aware that the answer space allocated is a guide to the length of the required response, and that marks are awarded for quality, not quantity; some candidates exceeded the space provided.

Specific Comments

Question 16

- (a) The better responses included a correct definition of a pathogen. However, a significant number of candidates were unable to do this.
- (b) Most responses provided correct distinguishing characteristics and features of the two microorganisms.

Question 17

This question was generally well answered, with the majority of candidates demonstrating a sound understanding of the concept of feedback.

Question 18

The better responses provided the main features of transcription and translation, using correct terminology. Weaker responses confused the processes with each other, or with other processes like translocation, transpiration and transportation.

Question 19

The better responses demonstrated a good basic understanding of their named disease by correctly describing all features in the table.

Question 20

The better responses related cause and effect correctly for each organism. Weaker responses simply stated a cause or an effect. Some candidates did not understand the concept of osmosis.

Question 21

- (a) The majority of responses correctly identified why excess carbon dioxide needs to be removed from the blood.
- (b) Most candidates stated at least one correct form in which carbon dioxide is transported in blood. In the better responses, two forms were stated.
- (c) Most candidates identified some features of an experimental method. A significant number of candidates confused reliability with accuracy or validity.

Question 22

This question was generally well answered, with the majority of candidates able to identify issues and provide points for and/or against tagging.

Question 23

- (a) This question was generally well answered.
- (b) This question was generally well answered.
- (c) In the better responses, cause and effect were related to show how crossing over affects the inheritance of genes. They also correctly identified where crossing over occurs.

Question 24

- (a) Most candidates represented the written description of the family by constructing a pedigree of some type. The better responses included a key, used conventional symbols and demonstrated an understanding of sex-linkage.
- (b) In the better responses, candidates made a correct prediction from their pedigree and supported their prediction.

Question 25

This question assessed Module 9.1 skills content and was generally well answered.

- (a) Most candidates were able to identify an appropriate method. A significant number of candidates named the type of data that could be collected without naming the method required to collect it.
- (b) Most candidates were able to state another valid piece of information.
- (c) In the better responses candidates clearly made judgements about the impacts which would result from the alternative strategies. A significant number of candidates did not support the claims they made, while some candidates simply identified impacts without making a judgement.

Question 26

In the better responses, candidates demonstrated a clear understanding of the adaptations of mangroves and related them to environmental conditions.

A significant number of candidates misinterpreted the term 'relative abundance' on the graph as referring to water, while others misinterpreted the terms 'high water mark' and 'low water mark' as meaning areas of high and low water availability.

Question 27

In the better responses, candidates explained the relationship between high mutation rates in the virus and the impact on the human immune response, and linked them to appropriate implications for both survival of the virus and humans.

The majority of candidates demonstrated a sound knowledge of the immune response. A considerable number of candidates confused terms such as antibodies, antigens and antibiotics, and many thought that antibiotics were used against viruses.

Section II – Options

Question 28 – Communication

- (a) (i) Most candidates were able to correctly identify the retina.
- (ii) Better responses used clear, well-labelled diagrams to show variation in the curvature of the lens, the condition of the ciliary muscles and suspensory ligaments and the refraction of the rays onto the retina for both near and far objects.
- (b) (i) The better responses gave characteristics and features for both conditions.
- (ii) Better responses clearly identified similarities and/or differences between both technologies, and often presented the response in table form.
- (c) Candidates were generally able to provide benefits and limitations of the two technologies. However, in a significant number of responses the judgement about the technologies was either limited or absent. Better responses often used a table to present points for and against the technologies and included clear examples to qualify the judgements.
- (d) (i) Most candidates were able to identify from the diagram two differences between the neurones.
- (ii) Most candidates were able to identify a benefit or limitation of the use of prepared slides or micrographs. In the better responses candidates made a clear judgement of the value of the use of slides or micrographs in gathering information about the structure of neurones.
- (iii) Many responses sketched in general terms the concept of ‘threshold’. Better responses were able to make the relationship between stimulus and action potential evident.

Question 29 – Biotechnology

- (a) (i) Most candidates were able to identify carbon dioxide as the second product.
- (ii) In the better responses, candidates included specific information about the features and characteristics of a modern fermentation process.
- (b) (i) Better responses clearly stated the meaning of ‘recombinant DNA technology’ and provided an example. A significant number of candidates were unable to provide an example.
- (ii) Most candidates were able to give characteristics and features of these processes. Better responses used a table to clearly identify similarities and/or differences.
- (c) Candidates were generally able to provide benefits and limitations for selected applications. However, in a significant number of responses the judgement was either absent or limited to a generalised judgement about biotechnology rather than to the case studies.

- (d) (i) This part of the question was generally well answered. Candidates are reminded that a supporting statement is required for a justification.
- (ii) Better responses provided detailed reasons for collecting a DNA sample from the victim.
- (iii) Many high quality responses included PCR techniques in amplifying small samples and the separation of fragments using electrophoresis.

Question 30 – Genetics: The Code Broken?

- (a) (i) This part of the question was generally well answered. A number of candidates confused the relative positions of phosphate and sugar molecules.
- (ii) This part of the question was generally well answered.
- (b) (i) The majority of candidates correctly identified both the number of chromosomes and the chromosomal disorder as trisomy. Some candidates incorrectly counted chromosomal pairs as single chromosomes.
- (ii) In the better responses, candidates clearly described the location of somatic and germ line mutations and compared the impact of each on the gene pool of a species.
- (c) Candidates were generally able to list benefits and limitations of the Human Genome Project. However, in a significant number of responses the judgement was either absent or limited to a generalised judgement of the project. Better responses provided points for and against the benefits and limitations and used specific examples to qualify the judgements.
- (d) (i) This part of the question was generally well answered.
- (ii) Better responses recognised gene linkage and provided both phenotypic and genotypic ratios.
- (iii) Better responses related the use of a test cross to finding the percentage of recombinant types. Some candidates confused cross-breeding with hybridisation and finding the relative position of genes with finding the exact location of genes. Candidates are reminded that diagrams can add value to the quality of a response.

Question 31 – The Human Story

- (a) (i) A significant number of candidates were unable to name karyotype analysis.
- (ii) (ii) This part of the question was generally well answered.
- (b) (i) This part of the question was generally well answered.
- (ii) Better responses identified the issues and developed points for and/or against the difficulty of interpreting the past using only fossil records. Weaker responses tended to list some of the difficulties.

- (c) In the better responses, candidates clearly linked the effects of population mobility, modern medicine and genetic engineering on human evolution, and gave an indication of possible future trends.
- (d)
 - (i) Most candidates correctly outlined an absolute dating technique. In the better responses, candidates were also able to clearly show the differences between relative and absolute dating.
 - (ii) This part of the question was generally well answered.
 - (iii) The better responses referred to the stimulus material to show how the models were similar or different.

Question 32 – Biochemistry

- (a)
 - (i) A significant number of candidates incorrectly named the pigment.
 - (ii) Better responses identified the link between the experimental set-up and photosynthesis and used this to name three pieces of information obtained from the experiments.
- (b)
 - (i) Most candidates were able to use the data to recognise the changes in oxygen production between plant X and Y. In the better responses, candidates related these differences to microscopic structural features such as size/number of chloroplasts.
 - (ii) The better responses identified the thylakoid membranes as the site of the light-dependent reactions and the stroma for the Calvin cycle, both located in the chloroplasts.
- (c) Candidates were generally able to identify issues and provide points for and/or against the advantages and disadvantages of renewable resources, derived from biomass, being used to replace non-renewable resources. However, in a significant number of responses the judgement was either absent or limited to a generalised judgement rather than to the specific examples.
- (d)
 - (i) This part of the question was generally well answered.
 - (ii) The better responses summarised the steps of photosystem I and II into a simple linked flow diagram and showed where the products of these reactions moved on to further parts of the photosynthetic pathway.

Biology

2005 HSC Examination Mapping Grid

Question	Marks	Content	Syllabus outcomes
Section I Part A			
1	1	9.2.2.21, 9.2.2.2.2	H6
2	1	9.3.4.3.5	H9
3	1	9.4.2.2.1, 9.4.6.2.2	H6
4	1	9.3.2.2.5, 9.3.3.2.2	H9
5	1	9.4.3.3.2	H6
6	1	9.4.6.1	H5
7	1	9.2.3.2.9, 9.2.3.3.7	H7, H14
8	1	9.4.2.3.2, 9.4.2.2.2	H8
9	1	9.2.1.3.2	H6, H12
10	1	9.3.3.2.3, 9.3.4.2.1	H9, H14
11	1	9.4.5.2.3	H6
12	1	9.2.3.2.4	H6, H14
13	1	9.2.2.3.2	H6, H12
14	1	9.3.3.2.6, 9.3.3.2.8	H9, H14
15	1	9.3.1.2.3	H10
Section I Part B			
16 (a)	1	9.4.2.2.3	H13
16 (b)	2	9.4.2.2.3,	H13
17	3	9.2.1.2.6, 9.2.1.2.5	H6
18	4	9.3.4.2.2/3, 9.3.4.3.1	H6, H13
19	4	9.4.3.3.3	H6, H12, H13
20	3	9.2.3.3.4	H6, H12, H14
21 (a)	1	9.2.2.2.5	H6
21 (b)	2	9.2.2.2.1	H6
21 (c)	5	9.2.2.3.1	H11, H12

Question	Marks	Content	Syllabus outcomes
22	4	9.4.7.3.3, 9.4.7.2.1, 9.4.7.3.2	H1, H4
23 (a)	1	9.3.3.3.1	H9, H12
23 (b)	1	9.3.3.3.1	H9, H12
23 (c)	3	9.3.3.3.1	H9
24 (a)	4	9.3.2.3.1	H13, H14
24 (b)	2	9.3.2.2.6	H14
25 (a)	1	9.1	H4, H11
25 (b)	2	9.1	H4, H11
25 (c)	5	9.1	H4, H10
26	4	9.2.3.2.8, 9.2.3.3.6	H7, H12, H14
27	8	9.3.4.2.4, 9.4.4.2.2, 9.4.5.3.1, 9.4.5.2.1	H6, H10
Section II			
Question 28 — Communication			
28 (a) (i)	1	9.5.2.3.1	H6
28 (a) (ii)	3	9.5.3.3.2	H6, H12
28 (b) (i)	2	9.5.3.2.5	H6, H14
28 (b) (ii)	4	9.5.3.3.3	H6, H12
28 (c)	7	9.5.6.3.3	H3, H5, H14
28 (d) (i)	2	9.5.7.3.1	H6, H14
28 (d) (ii)	3	9.5.7.3.1	H6, H14
28 (d) (iii)	3	9.5.7.3.1	H6, H14
Section II			
Question 29 — Biotechnology			
29 (a) (i)	1	9.6.2.2.1, 9.6.3.2.1	H8
29 (a) (ii)	3	9.6.3.3.1, 9.6.3.3.2	H8, H12
29 (b) (i)	2	9.6.5.2.1, 9.6.5.2.2	H7
29 (b) (ii)	4	9.6.1.3.1, 9.6.1.2.3, 9.6.5.3.3	H7, H14
29 (c)	7	9.6.6.2.2, 9.6.6.2.3, 9.6.6.2.4, 9.6.6.3.1	H5, H14
29 (d) (i)	2	9.6.6.2.1	H6, H12, H14
29 (d) (ii)	3	9.6.6.2.1	H6, H14

Question	Marks	Content	Syllabus outcomes
29 (d) (iii)	3	9.6.6.2.1	H6, H14
Section II			
Question 30 — Genetics: The Code Broken?			
30 (a) (i)	2	9.7.1.3.1	H9, H14
30 (a) (ii)	2	9.7.1.3.1	H9, H14
30 (b) (i)	2	9.7.6.2.1	H9, H14
30 (b) (ii)	4	9.7.6.2.4	H9, H14
30 (c)	7	9.7.4.2.1, 9.7.4.2.2	H3, H5
30 (d) (i)	1	9.7.3.2.2	H9, H14
30 (d) (ii)	3	9.7.2.3, 9.7.3.3.1	H9, H13, H14
30 (d) (iii)	4	9.7.3.3.4	H6, H9, H14
Section II			
Question 31 — The Human Story			
31 (a) (i)	1	9.8.2.3.1	H10, H14
31 (a) (ii)	3	9.8.2.3.1	H10, H14
31 (b) (i)	2	9.8.1.2.3	H10
31 (b) (ii)	4	9.8.2.2.5	H10
31 (c)	7	9.8.6.1	H5
31 (d) (i)	3	9.8.2.2.3	H10, H14
31 (d) (ii)	2	9.8.4.2.1	H10, H14
31 (d) (iii)	3	9.8.3.3.3	H10, H14
Section II			
Question 32 — Biochemistry			
32 (a) (i)	1	9.9.2.2.1, 9.9.3.3.4	H6
32 (a) (ii)	3	9.4.3	H6, H14
32 (b) (i)	3	9.9.8.3.1, 9.9.8.2.2	H6, H14
32 (b) (ii)	3	9.9.8.3.2	H6
32 (c)	7	(9.9.1)	H3, H5
32 (d) (i)	3	9.9.3.3.4	H6, H14
32 (d) (ii)	5	9.9.4.2.5, 9.9.4.3.2	H6, H14

2005 HSC Biology Marking Guidelines

Section I, Part B

Question 16 (a)

Outcomes assessed: H13

MARKING GUIDELINES

Criteria	Marks
• Correct definition of pathogen	1

Question 16 (b)

Outcomes assessed: H13

MARKING GUIDELINES

Criteria	Marks
• Characteristics for each organism that distinguishes them	2
• One characteristic of one of the organisms	1

Question 17

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
- Identifies the main features of a role of the nervous system using an example - Includes concepts of detection and response	3
Identifies a role of the nervous system using an example	2
- Makes a correct statement about the nervous system OR - States an example	1

Question 18

Outcomes assessed: H6, H13

MARKING GUIDELINES

Criteria	Marks
Names TWO processes correctly and indicates main features of the processes	4
- Names TWO processes correctly and indicates main feature of ONE process OR - Indicates main features of BOTH processes without naming	3
- Names ONE process correctly and indicates main feature of named process OR - ONE correct name and ONE correct function (not linked) OR - TWO correct outlines with correct names but in wrong position OR - TWO processes correctly named or outlined	2
- ONE process correctly named OR - ONE correct outline with correct name but in wrong position	1

Question 19

Outcomes assessed: H6, H12, H13

MARKING GUIDELINES

Criteria	Marks
• Provides characteristics of ALL the features related to the named infectious disease	4
• Provides characteristics of SOME features related to the named infectious disease	2–3
• Provides a characteristic related to the named disease	1

Question 20

Outcomes assessed: H6, H12, H14

MARKING GUIDELINES

Criteria	Marks
• Identifies, for each organism, cause and effect on urine output	3
• Identifies cause and effect for TWO organisms	2
• Identifies cause and effect for ONE organism OR • Makes a correct statement about water concentration in the relevant environment	1

Question 21 (a)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Identifies a reason for the removal of carbon dioxide	1

Question 21 (b)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• States TWO forms of carbon dioxide	2
• States ONE form of carbon dioxide	1

Question 21 (c)

Outcomes assessed: H11, H12

MARKING GUIDELINES

Criteria	Marks
Identifies the main features of the method for an appropriate investigation, stating appropriate variables and stating a method for checking reliability of data	4–5
Identifies some of the main features of the investigation with reference to the variables or reliability	2–3
Makes some correct statements relating to the method	1

Question 22

Outcomes assessed: H1, H4

MARKING GUIDELINES

Criteria	Marks
Identifies the issues and describes logical points for and/or against the use of this strategy in controlling the spread of disease	4
Explains the use of this strategy	2–3
Outlines the use of this strategy	1

Question 23 (a)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
Correctly identifies characteristic	1

Question 23 (b)

Outcomes assessed: H9, H12

MARKING GUIDELINES

Criteria	Marks
Identifies process	1

Question 23 (c)

Outcomes assessed: H9

MARKING GUIDELINES

Criteria	Marks
Identifies where crossing over occurs. Links cause and effect of crossing over	3
- Identifies crossing over AND describes crossing over OR the effect of crossing over OR - Describes crossing over AND the effect of crossing over	2
- Identifies where crossing over occurs OR - Describes crossing over OR - Describes the effect of crossing over	1

Question 24 (a)

Outcomes assessed: H13, H14

MARKING GUIDELINES

Criteria	Marks
- Constructs pedigree correctly showing affected and unaffected members. Anne identified as a carrier - Correct key included	4
- Constructs pedigree with some errors - Key incomplete	2–3
- Constructs the framework of the pedigree - No key	1

Question 24 (b)

Outcomes assessed: H14

MARKING GUIDELINES

Criteria	Marks
- Makes a prediction based on pedigree in part (a) - Supports prediction with logical argument	2
Makes a correct statement from pedigree drawn	1

Question 25 (a)

Outcomes assessed: H4, H11

MARKING GUIDELINES

Criteria	Marks
Names appropriate method	1

Question 25 (b)

Outcomes assessed: H4, H11

MARKING GUIDELINES

Criteria	Marks
States one valid/relevant piece of information directly applicable to the plague locust AND	2
Describes how it can be used to develop a method of control	
States one valid/relevant piece of information directly applicable to the plague locust	1

Question 25 (c)

Outcomes assessed: H4, H10

MARKING GUIDELINES

Criteria	Marks
Assesses issues from BOTH strategies that relate to society and the environment based on the issues identified	5
Discusses the impact of BOTH strategies on society AND the environment	4
Identifies the impact of ONE strategy on society AND the environment OR	3
Identifies the impact of BOTH strategies on either society OR the environment	
Identifies the impact of ONE strategy on either society OR the environment	2
States an impact of ONE strategy	1

Question 26

Outcomes assessed: H7, H12, H14

MARKING GUIDELINES

Criteria	Marks
- Correctly identifies each mangrove species on the graph - Gives reasons for the placement of BOTH species referring to the purpose of the specific adaptation of each species and relates this to the environmental conditions	4
- Correctly identifies each mangrove species on the graph - Describes the purpose of each adaptation OR - Gives reasons for the placement of one species, referring to the purpose of the specific adaptation for that species, and relates this to the environmental conditions	3
- Correctly identifies each mangrove species on the graph - Describes the purpose ONE adaptation OR - Describes two adaptations	2
- Correctly identifies each mangrove species OR - Describes one adaptation	1

Question 27*Outcomes assessed: H6, H10***MARKING GUIDELINES**

Criteria	Marks
- Indicates the progression from one mutation to another - Explains relationships between the TWO components including the immune response - Explains appropriate implications on both survival of the virus and human health	7–8
- Describes relationships between the TWO components - Describes appropriate implications on both survival of the virus AND human health	5–6
- Identifies a relationship between the components - Identifies appropriate points that impact on virus survival AND/OR human health	3–4
Identifies points appropriate to the theme	1–2

Section II – Option

Question 28 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
• Retina	1

Question 28 (a) (ii)

Outcomes assessed: H6, H12

MARKING GUIDELINES

Criteria	Marks
• Draws TWO correct labelled diagrams	3
• Draws TWO diagrams with some labels	2
• Draws ONE correct diagram	1

Question 28 (b) (i)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
• Provides characteristics and features for both conditions	2
• Provides characteristics and features for ONE condition OR • Outlines both conditions	1

Question 28 (b) (ii)

Outcomes assessed: H6, H12

MARKING GUIDELINES

Criteria	Marks
• Shows how BOTH technologies are similar or different using descriptions and/or examples	4
• Show how BOTH the technologies are similar or different by outlining these	2–3
• States one technology used for one condition	1

Question 28 (c)

Outcomes assessed: H3, H5, H14

MARKING GUIDELINES

Criteria	Marks
- Discusses benefits and limitations about both technologies - Provides a judgement about each technology based on the criteria used and uses examples in support	7
- Provides points for and against both technologies - Provides a judgement - Provides an example	5–6
Outlines a benefit and disadvantage for both technologies. An example is given for both	3–4
Identifies a benefit and/or disadvantage for a technology	1–2

Question 28 (d) (i)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
Two pieces of information from the diagram	2
One piece of information from the diagram	1

Question 28 (d) (ii)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
- Describes the value of a use of micrograph or prepared slide - Makes a judgement about the use of these for developing our understanding of neurones	3
- Outlines the use of micrograph/slide and describes how these help us understand structure of neurones - Makes a judgement	2
- States a use prepared slides/micrograph OR - Makes a judgement	1

Question 28 (d) (iii)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
Relates cause and effect of action potential and provides supporting evidence of when a stimulus does not elicit a response	3
Outlines an action potential and identifies a situation where a stimulus does not elicit a response	2
Makes a correct statement about an action potential	1

Question 29 (a) (i)

Outcomes assessed: H8

MARKING GUIDELINES

Criteria	Marks
Carbon dioxide	1

Question 29 (a) (ii)

Outcomes assessed: H8, H12

MARKING GUIDELINES

Criteria	Marks
Provides features and characteristics of a named modern industrial fermentation process	3
Outlines a modern fermentation process	2
Names a modern fermentation process	1

Question 29 (b) (i)

Outcomes assessed: H7

MARKING GUIDELINES

Criteria	Marks
- States meaning of the term recombinant DNA technology - Gives an appropriate example	2
- Correctly defines the term recombinant DNA technology OR - Gives an appropriate example	1

Question 29 (b) (ii)*Outcomes assessed: H7, H14***MARKING GUIDELINES**

Criteria	Marks
• Describes TWO or more similarities and/or differences between artificial selection and recombinant DNA technologies	4
• Describes at least ONE similarity and/or difference between artificial selection and recombinant DNA technology	2–3
• Identifies ONE similarity or difference between artificial selection and recombinant DNA technology	1

Question 29 (c)

Outcomes assessed: H5, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies appropriate issues and describes both limitations and benefits with points relevant to particular applications relating to TWO of the areas of biotechnology - Provides a judgement about each application based on the criteria used and uses examples in support	6–7
- Describes both limitations and benefits with points relevant to particular applications - Provides a judgement	4–5
- Describes a particular application of biotechnology research - Identifies limitations and benefits without links to the particular application	2–3
- Describes a particular application of biotechnology research OR - Identifies some limitations or benefits with reference to a particular application	1

Question 29 (d) (i)

Outcomes assessed: H6, H12, H14

MARKING GUIDELINES

Criteria	Marks
• Identifies suspect and gives reasons	2
• Identifies suspect and gives reason	1

Question 29 (d) (ii)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
- Provides reasons for the use of DNA profiles and the need to identify the victim's DNA in the evidence sample - Provides reasons for the collection of DNA from both the victim and suspect(s) - Link the DNA samples collected to eliminate the victim's DNA from the evidence sample	3
- Provides reasons for the collection of DNA from both the victim and suspect(s) OR - Gives a simple statement relating the victim's DNA and the DNA in the evidence sample	1–2

Question 29 (d) (iii)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
• Identifies main features and/or the uses of DNA analysis in forensic cases	2–3
• Identifies a use of DNA analysis	1

Question 30 (a) (i)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
• Identifies TWO parts	2
• Identifies ONE Part	1

Question 30 (a) (ii)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
• States TWO reasons	2
• States ONE reason	1

Question 30 (b) (i)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
• Both answers correct	2
• ONE answer correct	1

Question 30 (b) (ii)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
- Identifies where each mutation occurs - Describes the effects of each in terms of the species/population showing differences and/or similarities	4
- Identifies the cells where each mutation occurs - Describes the effect/ impact of one type of mutation on the species/population showing similarities and/or differences	2–3
Identifies in which cells one of the mutations occurs	1

Question 30 (c)

Outcomes assessed: H3, H5

MARKING GUIDELINES

Criteria	Marks
- Identifies appropriate issues of both limitations and benefits with relevant points for/and or against - Makes a judgement based on the criteria addressed	6–7
- Identifies appropriate issues of both limitations and benefits with relevant points for/and or against - Outlines links with relevant examples - Provides a judgement	4–5
- Identifies a benefit and issues associated, using examples OR - Identifies a limitation and issues associated using examples	2–3
Identifies one limitation or benefit	1

Question 30 (d) (i)

Outcomes assessed: H9, H14

MARKING GUIDELINES

Criteria	Marks
Correct ratio	1

Question 30 (d) (ii)

Outcomes assessed: H9, H13, H14

MARKING GUIDELINES

Criteria	Marks
- Punnett square – correctly constructed 3 genotypes and phenotypes correctly predicted	3
- Punnett square correct - Phenotypes correct – no genotypes ratio or vice versa OR - Correct phenotypes and genotypes ratio with no Punnett square	2
Punnett square correctly constructed with no ratios given	1

Question 30 (d) (iii)

Outcomes assessed: H6, H9, H14

MARKING GUIDELINES

Criteria	Marks
- Gives a reason for cross breeding experiments - Links cause and effect of phenotypes to parents - Relates position of linked genes to percentage of offspring	3–4
- Outlines features of cross-breeding OR - Identifies crossing over as the cause of recombinants	2
Makes a correct statement about cross-breeding or linked genes	1

Question 31 (a) (i)

Outcomes assessed: H10, H14

MARKING GUIDELINES

Criteria	Marks
Correct answer	1

Question 31 (a) (ii)

Outcomes assessed: H10, H14

MARKING GUIDELINES

Criteria	Marks
Lists THREE correct reasons	3
Lists TWO correct reasons	2
Names ONE correct reason	1

Question 31 (b) (i)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• TWO appropriate features correctly identified	2
• ONE appropriate feature correctly identified	1

Question 31 (b) (ii)

Outcomes assessed: H10

MARKING GUIDELINES

Criteria	Marks
• Identifies appropriate issues and describes relevant points for and/or against	4
• Describes the fossil record and/or how it may be used to interpret the past	2–3
• Makes a correct statement about the fossil record or fossils	1

Question 31 (c)

Outcomes assessed: H5

MARKING GUIDELINES

Criteria	Marks
- Describes logical relationships between human evolution and increased population mobility, modern medicine and genetic engineering AND - Relates these to future trends	7
Describes logical relationships between human evolution and increased population mobility, modern medicine and genetic engineering	5–6
Outlines the relationship between increased population mobility, and/or modern medicine and/or genetic engineering to human evolution	2–4
Identifies an appropriate issue relevant to the theme	1

Question 31 (d) (i)

Outcomes assessed: H10, H14

MARKING GUIDELINES

Criteria	Marks
- Distinguishes between relative and absolute dating AND - Outlines an appropriate absolute dating technique	3
- Distinguishes between relative and absolute dating OR - Outlines an appropriate absolute dating technique OR - Outlines one of 'relative' or 'absolute' and outlines an appropriate technique	2
- Provides an outline of 'relative' dating or 'absolute' dating OR - Identifies one technique	1

Question 31 (d) (ii)

Outcomes assessed: H10 H14

MARKING GUIDELINES

Criteria	Marks
Indicates a main feature of identified mechanism	2
Identifies a mechanism	1

Question 31 (d) (iii)

Outcomes assessed: H10, H14

MARKING GUIDELINES

Criteria	Marks
Shows similarities and/or differences in the evidence between TWO different named models of human evolution	3
Outlines ONE named model of human evolution	2
- Identifies ONE feature of ONE named model of human evolution OR - Identifies TWO models	1

Question 32 (a) (i)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Correctly names pigment	1

Question 32 (a) (ii)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
Identifies THREE correct facts	3
Identifies TWO correct facts	2
Identifies ONE correct fact	1

Question 32 (b) (i)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
Identifies TWO microscopic structural differences between plant X & Y and relates them to oxygen production	3
Identifies TWO microscopic structural differences based on the data provided	2
Identifies two microscopic characteristics of plants associated with photosynthesis	1

Question 32 (b) (ii)

Outcomes assessed: H6

MARKING GUIDELINES

Criteria	Marks
Identifies both sites correctly and outlines the location of the appropriate reaction	3
Names both sites OR	2
Names one site and makes an attempt at an explanation	
Names a site for photosynthesis in any part of the reaction	1

Question 32 (c)

Outcomes assessed: H5, H3

MARKING GUIDELINES

Criteria	Marks
Describes the production of three named materials using new technologies and makes a judgement about their replacement with the use of photosynthesis	6–7
- Describes at least TWO named materials using new technologies - Identifies links with use of photosynthesis and replacement of current materials - Discusses their use as replacement for current materials - Makes a judgement	4–5
- Identifies appropriate potential uses of photosynthesis - Links use of photosynthesis and replacement of existing material(s)	2–3
- Identifies an appropriate use of photosynthesis OR - Identifies an appropriate material obtained from other non-renewable resources	1

Question 32 (d) (i)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
- Relates the absorption of light to the production of chemical energy required by the plant to grow - Demonstrates the relationship between maximum light penetration to maximising plant growth	3
Outlines function of appropriate filter	2
Identifies an appropriate filter	1

Question 32 (d) (ii)

Outcomes assessed: H6, H14

MARKING GUIDELINES

Criteria	Marks
- Provides characteristics and features of PSI and PSII, identifying the site and products of each - Provides cause and effect of the processes that occur	4–5
Outlines the role of PSI and PSII and outlines the processes that occur in each	2–3
Identifies the role of PSI or PSII	1